

HW 2 Part 2 - Drug Deaths

1) Getting started

- Import the data from the csv file.
- Investigate it using pandas' `info` command.
- What is the data type of the Date column?

```
In [1]: import pandas as pd
df = pd.read_csv('drugdeaths.csv')
df.head()
```

Out[1]:

	CaseNumber	Date	Sex	Race	Age	Residence City	Residence State	Residence County	
0	13-16336	11/9/13	Female	White	53.0	GROTON	NaN	NEW LONDON	
1	12-18447	12/29/12	Male	White	30.0	WOLCOTT	NaN	NEW HAVEN	W
2	14-2758	2/18/14	Male	White	43.0	ENFIELD	NaN	NaN	
3	14-13497	9/7/14	Female	White	24.0	WALLINGFORD	NaN	NaN	WALI
4	13-14421	10/4/13	Female	White	26.0	WEST HAVEN	NaN	NEW HAVEN	WE

5 rows × 32 columns



```
In [2]: df.info()
```

```

<class 'pandas.DataFrame'>
RangeIndex: 3581 entries, 0 to 3580
Data columns (total 32 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   CaseNumber                           3581 non-null   str
1   Date                                3581 non-null   str
2   Sex                                  3579 non-null   str
3   Race                                3572 non-null   str
4   Age                                  3580 non-null   float64
5   Residence City                       3490 non-null   str
6   Residence State                      1663 non-null   str
7   Residence County                    2885 non-null   str
8   Death City                          3579 non-null   str
9   Death State                         1702 non-null   str
10  Death County                        2405 non-null   str
11  Location                            3561 non-null   str
12  DescriptionofInjury                 998 non-null    str
13  InjuryPlace                        3504 non-null   str
14  ImmediateCauseA                    3579 non-null   str
15  Heroin                             1936 non-null   str
16  Cocaine                             997 non-null    str
17  Fentanyl                           1114 non-null   str
18  Oxycodone                          491 non-null    str
19  Oxymorphone                        95 non-null     str
20  EtOH                               772 non-null    str
21  Hydrocodone                        98 non-null     str
22  Benzodiazepine                     884 non-null    str
23  Methadone                          336 non-null    str
24  Amphet                             81 non-null     str
25  Tramad                             79 non-null     str
26  Morphine (not heroin)               50 non-null     str
27  Other                              320 non-null    str
28  Any Opioid                         710 non-null    str
29  MannerofDeath                     3575 non-null   str
30  AmendedMannerofDeath               538 non-null    str
31  DeathLoc                           3581 non-null   str
dtypes: float64(1), str(31)
memory usage: 895.4 KB

```

The data column has `string` data type.

2) Reimport the data but this time add the argument `parse_dates = ['Date']`.

Has the data type of the Date column changed? If so, to what?

```
In [3]: df = pd.read_csv('drugdeaths.csv', parse_dates = ['Date'])
```

```
C:\Users\ifrommer\AppData\Local\Temp\ipykernel_26780\2247979404.py:1: UserWarning: Could not infer format, so each element will be parsed individually, falling back to `dateutil`. To ensure parsing is consistent and as-expected, please specify a format.
```

```
df = pd.read_csv('drugdeaths.csv', parse_dates = ['Date'])
```

Below is a snippet of `df.info()`. We can see that the `Date` column changed from type `string` to `datetime64[us]`

```
---  ---
0   CaseNumber      3581 non-null   str
1   Date          3581 non-null   datetime64[us]
```

3) Explain how you think the `parse_dates` argument works.

Answer: it tells `pandas` to process the column(s) passed to it as dates, converting to a `datetime` data type for easy processing as dates/times.

4) The command `df['Date'].dt.month` will return a Series with the month number of the corresponding date.

Use this to create a new column in the DataFrame named "Month" and show that it was created.

We will show a snippet of the DataFrame showing the first few rows and a sample of columns including *Date* and the new *Month* column.

```
In [25]: df['Month'] = df['Date'].dt.month
df.iloc[0:4][['Date', 'Age', 'Residence City', 'Death City', 'MannerofDeath', 'Month']]
```

```
Out[25]:
```

	Date	Age	Residence City	Death City	MannerofDeath	Month
0	2013-11-09	53.0	GROTON	GROTON	Accident	11
1	2012-12-29	30.0	WOLCOTT	WATERBURY	Accident	12
2	2014-02-18	43.0	ENFIELD	ENFIELD	Accident	2
3	2014-09-07	24.0	WALLINGFORD	WALLINGFORD	Accident	9

5) How many accidental drug deaths are there for each month in this dataset?

Display your answer sorted by month order.

Hint: use the "MannerofDeath" column.

```
In [26]: df['MannerofDeath'].value_counts()
```

```
Out[26]: MannerofDeath
Accident    3550
Pending      14
accident      9
Natural       1
ACCIDENT      1
Name: count, dtype: int64
```

Since we see multiple ways accident is written, but capture each one with `isin`.

```
In [27]: df_accs = df[df['MannerofDeath'].isin(['Accident', 'ACCIDENT', 'accident'])]
```

```
In [28]: df_accs['Month'].value_counts()
```

```
Out[28]: Month
6      350
3      327
2      326
5      324
1      319
4      314
11     303
10     288
12     275
7      249
9      248
8      237
Name: count, dtype: int64
```

sort in month order -

```
In [29]: df_accs['Month'].value_counts().sort_index()
```

```
Out[29]: Month
1      319
2      326
3      327
4      314
5      324
6      350
7      249
8      237
9      248
10     288
11     303
12     275
Name: count, dtype: int64
```

6) What is the city with the most accidental deaths in this dataset? How many deaths did this city have?

```
In [30]: df_accs['Death City'].value_counts()
```

```
Out[30]: Death City
HARTFORD      364
NEW HAVEN     262
WATERBURY     251
BRIDGEPORT    226
NEW BRITAIN   151
...
KENT          1
HAMPTON       1
BOLTON        1
NEW PRESTON   1
CORNWALL BRIDGE 1
Name: count, Length: 203, dtype: int64
```

```
In [31]: df_accs['Death City'].value_counts().index[0]
```

```
Out[31]: 'HARTFORD'
```

7) Find the average age of accidental deaths for all cities in the dataset.

Hint: start by using `groupby`

a. What city has the oldest average age of accidental deaths? What is the average age of accidental deaths for this city?

b. What city has the youngest average age of accidental deaths? What is the average age of accidental deaths for this city?

```
In [32]: df_dc = df_accs.groupby(['Death City'])
dc_mean_age = df_dc['Age'].mean()
```

```
In [33]: mean_age_sorted = dc_mean_age.sort_values()
mean_age_sorted
```

```
Out[33]: Death City
NEW PRESTON      17.0
STORRS           18.0
WARREN           21.0
EAST HARTLAND    21.0
WILTON           22.0
...
QUAKER HILL      58.0
SOUTHBURY        59.0
PLYMOUTH         60.0
VERNON-ROCKVILLE 62.0
GALES FERRY      65.0
Name: Age, Length: 203, dtype: float64
```

```
In [34]: mean_age_sorted[:1]
```

```
Out[34]: Death City
NEW PRESTON    17.0
Name: Age, dtype: float64
```

```
In [35]: mean_age_sorted[-1:]
```

```
Out[35]: Death City
GALES FERRY    65.0
Name: Age, dtype: float64
```

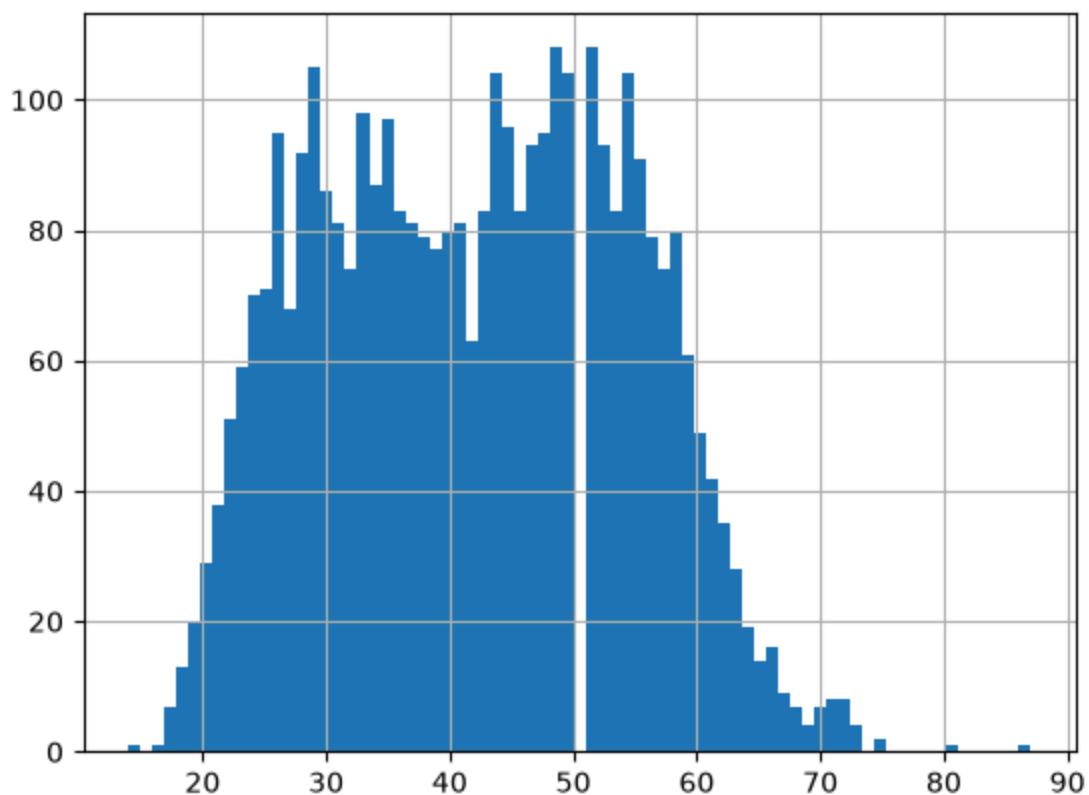
8) Generate a bar chart to display the frequency of the ages that occur in this dataset.

Note: this means for all ages found in the data set (not the city mean ages) and over all types of deaths (not just accidental)

Hint: consider using the hist command and setting the number of bins using trial and error.

```
In [36]: df['Age'].hist(bins=75)
```

```
Out[36]: <Axes: >
```



```
In [ ]:
```